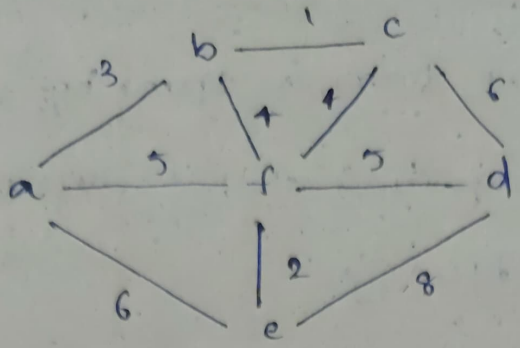
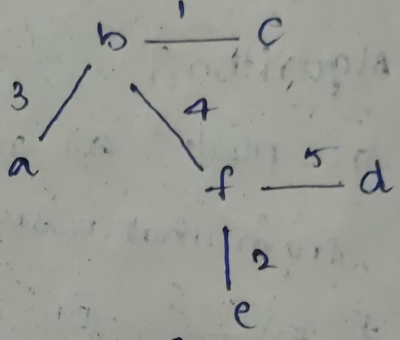
Tut-10

1.



a) Kruskals algorithm: smallest edge weights

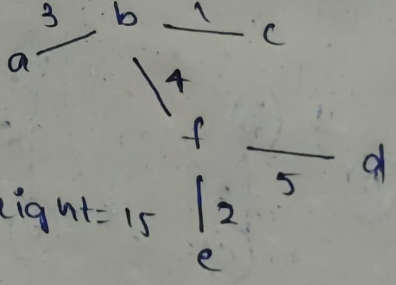
bc = 1
fe = 2
ab = 3
bf = 4
cf = 4
af = 5
fd = 5



weight = 15

b) prim's: start from a

ab = 3
bc = 1
bf = 4
fe = 2
fd = 5



weight = 15

2. a) True

min weight edge is always be a part of a MST

b) False

max weight edge never part of a MST

c) True

if distinct weights are present we will have only one possible MST.

3) a) G → undirected edge weighted graph

a) True

b) False
max weight edge might also be required

c) False
unique MST only in case of unique weights

4) True.
every minimum weight edge is part of at least ones MST.

5) G with weighted edges MST(T) of G.

a) update MST T when weight of single edge e is increased do nothing

Case i :- edge not in MST
do nothing

Case ii :- Edge in MST

1. Remove edge → split tree into 2 components
2. find min edge connecting the two components
3. add it

b) update MST T when weight of single edge e = decreases

Case i :- edge already in MST
do nothing.

Case ii :- Edge not in MST

1. Add edge → creates cycle
2. Remove max weight edge in that cycle.

6) 1. sort all edges in decreasing order of weight
2. initialize:

$T = \phi$

3. for each edge (u,v) in sorted order:
   if adding (u,v) does not form a cycle
   add it to T

4. stop when
   T has $(V-1)$ edges

$T_c = O(E \log V)$

7) a) False.
MST → minimizes total weight of the graph not pairwise distances.

b) False
with / without distinct weights → yet MST remains same it doesn't minimise pairwise distances.

8) a) True.
If unique edge weights are present, there exists only unique MST either first best / 2nd best.

b) Algorithm:

compute MST T (using kruskal/prim)

for every edge $e \notin T$

add e to T

// creates a cycle

find maximum weight edge f in that cycle

form new tree

$$T' = T + e - f$$

compute weight

$$cost(T') = cost(T) + w(e) - w(f)$$

$TC = O(E \log V)$